

Wookyoung Kim

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Daejeon, KR

20 papers SCI + KCI
peer-reviewed

18 patents domestic + 1
US

8 programs registered
software

5 transfers technology
transfers

Summary

Ph.D. thermal engineer with 10+ years in electronics cooling and two-phase heat transfer, from pulsating heat pipes (KAIST) to AI data center cooling (DLC, immersion, jet impingement) as PI/project lead at KIMM, and Associate Professor at the University of Science and Technology (UST), KIMM School. Combines hands-on experimental expertise (cryogenic testing, hydrogen heat exchangers, heat pumps) with production-grade software development (Python / FastAPI / React Native).

Education

Ph.D. · KAIST (Korea Advanced Institute of Science and Technology) · Mechanical Engineering 2017 – 2021

- Dissertation: Study on a Thermal Network-based Model for Predicting the Thermal Resistance of Pulsating Heat Pipes
- Advisor: Prof. Sung Jin Kim

M.S. · KAIST · Mechanical Engineering 2015 – 2017

- Thesis: Effect of artificial cavities on the thermal performance of pulsating heat pipes
- Advisor: Prof. Sung Jin Kim

B.S. · KAIST · Mechanical Engineering 2011 – 2015

Experience

2021 – Present

Senior Researcher · Korea Institute of Machinery and Materials (KIMM) · Daejeon, Republic of Korea 2021 – Present

Heat Pump Research Center, Department of Thermal Energy Solutions, Innovative Energy Machinery Research Division

- PI: Jet-enhanced immersion cooling for next-gen data center cooling
- PI: KIMMPROP - Thermophysical property iOS/Android app development
- Lead: PCHE design & testing for liquid hydrogen vaporizers (nitrogen tests to -185°C, 21 MPa; -220°C, 100 MPa target)
- Lead: Immersion cooling waste heat utilization and thermal management
- 20 journal papers (12 SCI + 8 KCI), 18 domestic patents, 1 U.S. patent, 8 registered software programs, 5 technology transfers

Associate Professor · University of Science and Technology (UST) · Daejeon, Republic of Korea 2026 – Present

KIMM School, Mechanical Engineering Systems

- Full-time faculty (Associate Professor), appointed term Sep 2026 - Aug 2031
- Graduate teaching and M.S./Ph.D. advising in thermal and energy systems

Publications

12 SCI · 8 KCI · 11 Conference

SCI / International Journal Articles 12 papers

2026	Vapor–liquid equilibrium measurements and mixture-model parametrization for the binary refrigerant systems R1234yf + CF3I and R1123 + CF3I <i>International Journal of Refrigeration</i> Kim, Dong Ho and Kim, Wookyoung and Lee, Kong Hoon
2026	Falling film evaporation of ammonia on smooth and Low-Fin tube arrays <i>Thermal Science and Engineering Progress</i> Kim, Jin Sub and Kim, Wookyoung and Kim, Hak Soo and Kim, Young
2025	Experimental and theoretical investigation of freezing phenomenon in Printed Circuit Heat Exchanger <i>Applied Thermal Engineering</i> Kim, Bogyom and Lee, Kong Hoon and Kim, Minchang and Moon, Sunyoung and Yoo, Jin Woo and Kim, Wookyoung
2024	Experimental studies on the VLE of R-32/R-125 and verification of experimental setup <i>Journal of Mechanical Science and Technology</i> Kim, Dong Ho and Kim, Wookyoung and Kim, Jeong Hyun and Kim, Hak Soo and Lee, Kong Hoon
2024	A numerical study on the performance of chemisorption heat pump according to various design conditions <i>Applied Thermal Engineering</i> Kim, Hak Soo and Kim, Dong Ho and Kim, Jin Sub and Kim, Wookyoung and Kim, Young
2024	Pool boiling heat transfer of ammonia outside enhanced tubes with various fin structures <i>Applied Thermal Engineering</i> Kim, Jin Sub and Kim, Wookyoung and Kim, Hak Soo and Kim, Young
2024	Experimental study of a chemisorption heat pump under different operation conditions <i>Applied Thermal Engineering</i> Kim, Hak Soo and Kim, Jeong Hyun and Kim, Jin Sub and Kim, Wookyoung and Kim, Young
2023	Characterization of liquid behavior in distributor of falling-film evaporator <i>Physics of Fluids</i> Kim, Jungchul and Kim, Chan Geun and Kim, Hak Soo and Kim, Wookyoung and Kim, Dong Ho
2023	Pool boiling heat transfer of ammonia outside a single tube with fin structures: Hysteresis phenomena and boiling enhancement <i>International Communications in Heat and Mass Transfer</i> Kim, Jin Sub and Kim, Wookyoung and Kim, Hak Soo and Kim, Young and Yoon, Seok Ho
2021	Fundamental issues and technical problems about pulsating heat pipes <i>Journal of Heat Transfer - Transactions of the ASME</i> Kim, Wookyoung and Kim, Sung Jin
2020	Effect of a flow behavior on the thermal performance of closed-loop and closed-end pulsating heat pipes <i>International Journal of Heat and Mass Transfer</i> Kim, Wookyoung and Kim, Sung Jin
2018	Effect of reentrant cavities on the thermal performance of a pulsating heat pipe <i>Applied Thermal Engineering</i> Kim, Wookyoung and Kim, Sung Jin
KCI / Korean Journal Articles 8 papers	
2026	A Study on the Design of an Air-to-Air Plate Heat Exchanger for 300°C High Temperature Heat Pumps <i>Korean Journal of Air-Conditioning and Refrigeration Engineering</i> Kim, Jin Sub and Shin, Dong Hwan and Shim, Jaechwan and Lee, Kong Hoon and Sohn, Sangho and Kim, Wookyoung and Song, Chanhoo
2024	Experimental Investigation on the Freezing Condition of Printed Circuit Heat Exchanger for Cryogenic Liquid Hydrogen Vaporizer <i>Journal of Hydrogen and New Energy</i> Kim, Wookyoung and Kim, Bokyem and Sohn, Sangho and Lee, Kong Hoon and Kim, Jungchul

2024	An Experimental Study on The Thermal Performance Measurement and Flow Visualization of Falling Film Evaporator Using R-1233ZD(E) Refrigerant <i>Korean Journal of Air-Conditioning and Refrigeration Engineering</i> Kim, Wookyoung and Kim, Hak Soo and Kim, Jungchul and Kim, Dong Ho
2023	An Experimental Study on the Performance Characteristics of a Falling Film Evaporator for R-1234ze(E) Refrigerant <i>Korean Journal of Air-Conditioning and Refrigeration Engineering</i> Kim, Dong Ho and Kim, Hak Soo and Kim, Jungchul and Kim, Wookyoung
2023	Comparative Study of Air Cooling and Immersion Cooling for the Thermal Management of a Cylindrical Battery Pack <i>Transactions of the Korean Society of Mechanical Engineers B</i> Kim, Jin Sub and Kim, Seul Ah and Shin, Dong Hwan and Kim, Wookyoung and Moon, Sunyoung and Chung, Yoong and Sohn, Sangho
2022	A Study on Anti-Icing Design by Conjugate Heat Transfer Analysis in a Lab-Scale Printed Circuit Heat Exchanger for Supply of Cryogenic High Pressure Liquid Hydrogen <i>Transactions of the Korean Hydrogen and New Energy Society</i> Sohn, Sangho and Kim, Wookyoung
2022	Numerical Study on the Performance of Hybrid Falling Film Evaporator <i>Korean Journal of Air-Conditioning and Refrigeration Engineering</i> Kim, Hak Soo and Kim, Wookyoung and Lee, Kong Hoon and Kim, Dong Ho
2022	Selection of the Equation of State for Thermodynamic Properties in the Low GWP Mixture Refrigerants <i>Transactions of the Korean Society of Mechanical Engineers B</i> Lee, Kong Hoon and Kim, Dong Ho and Kim, Wookyoung and Kim, Jin Sub and Kim, Young and Yoo, Jin Woo

Conference Proceedings 11 papers

2024	Experimental and theoretical investigation on avoiding freezing phenomena of Printed Circuit Heat Exchanger for cryogenic liquid hydrogen vaporizer <i>Proceedings of the 29th International Cryogenic Engineering Conference (ICEC29)</i> Kim, Wookyoung and others
2024	Understanding the thermal characteristics of falling film evaporation of R-1233ZD(E) refrigerant using flow visualization and heat transfer measurement <i>Proceedings of the 11th Asian Conference on Refrigeration and Air-conditioning (ACRA)</i> Kim, Wookyoung and others
2023	Experimental investigation on the flow and thermal characteristics of falling film evaporator using R-1233ZD(E) refrigerant <i>Proceedings of the 11th International Conference on Boiling and Condensation Heat Transfer (ICBCHT)</i> Kim, Wookyoung and others
2019	Experimental Investigation on the Thermal Performance of PHP Operating in a Circulation/oscillation Mode <i>Proceedings of the Korea-China-Japan Joint Symposium</i> Kim, Wookyoung and Kim, Sung Jin
2019	Comparison of thermal performance between CEPHP and CLPHP <i>Proceedings of the 4th Thermal and Fluids Engineering Conference (TFEC)</i> Kim, Wookyoung and Kim, Sung Jin
2018	Experimental investigation on the thermal performance of double-condenser pulsating heat pipes <i>Proceedings of the 10th International Conference on Boiling and Condensation Heat Transfer (ICBCHT)</i> Kim, Wookyoung and Kim, Sung Jin
2018	Development of deployable radiator for spacecraft using a pulsating heat pipe <i>Proceedings of the KSME 2018 Spring Annual Meeting (Thermal Engineering Division)</i> Kim, Wookyoung and Kim, Sung Jin

2018	Effects of circulating flow on the thermal performance of a pulsating heat pipe <i>Proceedings of the KSME 2018 Annual Meeting</i> Kim, Wookyoung and Kim, Sung Jin
2017	Experimental Investigation on the Thermal Performance of a Pulsating Heat Pipe with Artificial Cavities <i>Proceedings of the 1st Asian Conference on Thermal Sciences (ACTS 2017)</i> Kim, Wookyoung and Kim, Sung Jin
2017	Experimental investigation of artificial cavities on the thermal performance of a pulsating heat pipe <i>Proceedings of the ASME InterPACK 2017</i> Kim, Wookyoung and Kim, Sung Jin
2016	Effects of artificial cavities on the startup performance of a pulsating heat pipe <i>Proceedings of the KSME 2016 Annual Meeting</i> Kim, Wookyoung and Kim, Sung Jin

Research Projects 25 entries

Development of an AI-based Automated Design Tool for Data Center Direct Liquid Cooling (DLC) Systems 2026

AI-driven automated design tool for data center direct liquid cooling (DLC) systems, using surrogate modeling and optimization for cold-plate and cooling-system design to shorten thermal design iterations (PI)

Development of a Direct Liquid Cooling (DLC) System for Reducing Power Consumption in Data Centers 2026 – 2029

National R&D project (KETEP, Ministry of Climate, Energy and Environment) to develop a direct liquid cooling (DLC) system that reduces data center power consumption (Lead)

Jet-Enhanced Immersion Cooling for Data Centers 2025

Development of high-efficiency jet-impingement immersion cooling technology for next-generation high-heat-density data center servers (PI)

KIMMPROP - Thermophysical Property iOS/Android App 2025

Cross-platform iOS/Android app for thermophysical property calculations, built with React Native on CoolProp/REFPROP compiled to WASM; published on the App Store and Google Play (PI)

PCHE for Liquid Hydrogen Supply System 2021 – 2026

Core material development for liquid hydrogen supply system - PCHE performance testing and design, including anti-freezing channel design with conjugate heat transfer analysis and freezing-condition experiments (Lead)

Compact PCHE Design for Liquid Hydrogen 2022 – 2026

Compact heat exchanger (PCHE) design technology for liquid hydrogen below -200°C, 100 MPa class (Lead)

Immersion Cooling Waste Heat Utilization 2024 – 2028

Immersion cooling waste heat utilization and active thermal management technology for data centers (Lead)

300°C High-Temperature Heat Pump System 2023 – 2028

300°C-class high-temperature heat pump system for fossil fuel replacement in industrial processes (Participant)

Chemisorption Heat Pump with Electrochemical Compressor 2021 – 2025

Chemisorption heat pump system using electrochemical compressor (Participant)

Next-Gen Alternative Refrigerant & HVAC Technology 2021 – 2025

Next-generation alternative refrigerant and high-efficiency HVAC core technology development (Participant)

Low-Charge Industrial Heat Pump	2022 – 2023
Industrial heat pump with low refrigerant charge and simultaneous hot/cold water production (Participant)	
Water-Source Heating/Cooling Hybrid System	2022 – 2023
Water-source heating/cooling and regenerative heat hybrid system (Participant)	
Heat Pipe HX Performance Enhancement	2022
Heat pipe heat exchanger performance enhancement using boiling (Participant)	
Post H-class Gas Turbine Technology	2022
Post H-class gas turbine technology development (Participant)	
Compact Inverter Compressor Dress Care System	2022
Compact inverter compressor-based dress care system development; technology transferred to industry (Participant)	
Ultra-Thin Vapor Chamber Performance Enhancement	2022 – 2023
Performance enhancement technology for ultra-thin vapor chambers for semiconductor cooling; technology transferred to an industry partner (Participant)	
PCHE Freezing Test for Vehicle Heat Exchangers	2025
PCHE freezing performance test for automotive heat exchanger applications (PI)	
BTMS Integrated A/Con System Evaluation	2025
Battery thermal management system integrated A/Con system real-vehicle evaluation process development (Participant)	
3D Multiporous Cooling System for Ultra-high Heat Flux Applications	2021
Development of a 3D multi-porous cooling device for ultra-high-heat-flux heat-generating elements (Graduate Researcher)	
Smart protective suit with active cooling/ventilation and antivirus suit changing system	2020
Active cooling and ventilation system for a smart protective suit enabling on-body thermal management (Graduate Researcher)	
Next-Generation CPI Reliability Structural & Thermal Design	2020
Structural and thermal design technology to secure the reliability of next-generation CPI; NRF original-technology development project (Graduate Researcher)	
OHP High-Pressure Working-Fluid Charging Technology	2020
Technical service for high-pressure working-fluid charging of oscillating heat pipes (OHP); commissioned by Korea Institute of Energy Research (Graduate Researcher)	
Thermal optimization of MMU rear housing for reducing the weight	2019 – 2020
Heat-dissipation performance optimization and weight reduction of an MMU rear housing; industry project funded by Samsung Electronics (Graduate Researcher)	
Development of a Flexible Thin-Film Ultra-Thermal-Conductor	2015 – 2020
NRF basic research on flexible thin-film ultra-high-thermal-conductivity heat spreaders based on pulsating heat pipes (Graduate Researcher)	
Study on a smart space thermal management system	2016 – 2018
Pulsating-heat-pipe-based deployable radiator and smart thermal control device for spacecraft; contract with Korea Aerospace University (Graduate Researcher)	

Patents & Registered Software

Patents 19 registered

2023	Heat exchanger with anti-freezing capability (1)
2023	Heat exchanger with anti-freezing capability (2)
2024	Modular fin-tube reactor
2024	Fin-tube type reactor
2022	Tray structure for heat exchanger
2022	Oil recovery system for refrigeration systems (1)
2022	Oil recovery system for refrigeration systems (2)
2023	Micro-channel reactor
2024	Immersion cooling device
2023	Geothermal heat supply device and heating system
2023	Heat exchanger thermal performance testing apparatus
2022	Ternary refrigerant composition and heat pump system
2023	Display device
2022	Duct assembly and combustor
2022	Immersion cooling HVAC system and method
2022	Heat pipe integrated reactor for adsorption heat pump
2023	Adsorption heat pump evaporator and system
2018	Plate pulsating heat spreader with artificial cavities
2019	Plate pulsating heat spreader with artificial cavities (US 10,436,520 B2)

Registered Software 8 programs

2024	PCHE thickness design program (C-2024-052179)
2024	High-temperature heat pump cycle design program (C-2024-038691)
2023	Heat exchanger HTC uncertainty analysis program (C-2023-059071)
2023	Lab-scale heat exchanger analysis program (C-2023-056463)
2025	100 kg/hr hydrogen PCHE design program (C-2025-056891)
2025	Adsorption cooling simulation program (C-2025-034652)
2024	Vapor chamber performance analysis program (C-2024-041200)
2024	Fin-tube heat exchanger design program (C-2024-038692)

Core Skills

EXPERIMENTAL	Two-phase heat transfer testing · Cryogenic testing (nitrogen, down to -185°C) · High-pressure testing (up to 21 MPa) · PCHE diffusion bonding & fabrication · Immersion cooling (CPU/GPU servers) · Flow visualization (high-speed camera) · VLE measurement & gas chromatography
ANALYTICAL & COMPUTATIONAL	Thermal network modeling · Heat exchanger design (PCHE, S&T, PHE) · CFD simulation · Surrogate modeling & design optimization · CoolProp/REFPROP
SOFTWARE DEVELOPMENT	Python, JavaScript, TypeScript · FastAPI, React Native · Git/GitHub, Linux, CMake · Cross-platform app development (iOS/Android) · AI/LLM agent development

Research Interests

Data Center Cooling

Immersion cooling · Direct liquid cooling · Jet impingement · PUE optimization

Heat Exchangers

PCHE · Shell & Tube · Plate HX · Cryogenic applications

Low GWP Refrigerant Systems

Eco-friendly refrigerants · VLE measurement · Equation of state development

Cryogenic Heat Transfer

Liquid hydrogen vaporizer · Supercritical fluid · Sub-200°C systems

High-Heat-Flux Electronics Cooling

CPU/GPU cold plate · Focused cooling · Next-gen chip thermal management

Heat Pump Systems

Chemisorption · Adsorption · High-temperature (300°C)

Two-Phase Heat Transfer

Boiling · Evaporation · Pulsating heat pipes · Vapor chambers

Professional Affiliations

Council Member · Korean Society of Mechanical Engineers (KSME)	2026-04 – Present
Committee Member - Thermal Management / Cooling · SAREK Data Center Facility Technology Division Committee	2026-03 – Present
Regular Member · Society of Air-Conditioning and Refrigerating Engineers of Korea (SAREK)	2022-04 – Present
Regular Member · Korean Society of Mechanical Engineers (KSME)	2021-05 – Present